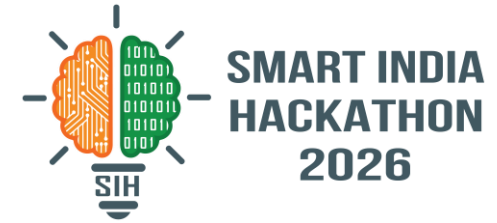


SMART INDIA HACKATHON 2026

TITLE PAGE



MOLE · Mine Observation & Live-alert Engine

Problem Statement ID – SIH26025

Problem Statement Title – Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India

Theme – Disaster Management

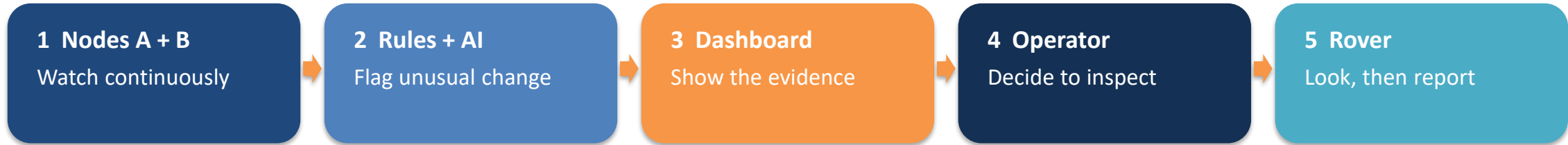
PS Category – Hardware

Team ID – To be assigned on the SIH portal after college shortlist

Team Name – WE COOK



❖ Proposed solution · how it addresses SIH26025 · what is unique



Isolation Forest is early anomaly detection — a precursor stage toward prediction. It does not output a collapse probability.

What MOLE does for SIH26025

- Node A watches surface tilt vs a captured baseline.
- Node B watches vibration / disturbance at a second point.
- Watch / Alert latch with a written reason and history.
- Rover inspects only after the operator decides — it is not the monitor.
- One local dashboard. Rover vibration is never treated as Node B.

Why this is not a copy of other student kits

- Walk-in checks send a person in first. Nodes keep watching.
- InSAR / surveys are slow for a heading. This is a live tabletop loop.
- Rover-only kits have no standing surface watch.
- Unexplained AI scores hide the reason. Rules and AI are shown separately.

Scientific basis (proxy, not a geophone)

- MPU6050 tilt and vibration are low-cost proxies for mounting-surface micro-movement. They are not professional microseismic arrays.
- Published Indian longwall work: microseismic event-rate changes can precede roof / ground failure (Sivakumar et al., 2005, Rajendra / SECL).
- DGMS (S&T) Tech. Circular 01 of 2017, cl. 6.1 asks longwall mines for surface geophones. That is field-grade method — not this tabletop prototype.

Stakes: CIL recorded 22 fatal accidents and 24 fatalities in 2024 up to November (MoC Annual Report 2024-25, Ch.14, Table 2; subject to DGMS reconciliation). Strata / roof control remains a named underground safety focus.

- Technologies · methodology · working prototype

Hardware (tabletop prototype)

- Node A: ESP32-WROOM-32 + MPU6050 for tilt / movement.
- Node B: ESP32-WROOM-32 + MPU6050 for vibration.
- Rover: motors + driver, remote FWD / REV / LEFT / RIGHT / STOP.
- USB ESP32 gateway: radio/serial infrastructure, not a fourth node.
- Parts are already with the team. This is a demonstration rig, not mine-certified gear.

Software (one laptop)

- Python dashboard. SQLite event history.
- Rules: Watch / Alert after 3 consecutive samples; 5 s stale → UNKNOWN.
- AI: scikit-learn Isolation Forest vs learned normal. Anomaly detector only.
- AI cannot clear a rule alert. If untrained, show INACTIVE.
- Every reading is source-tagged and timed.

Communication

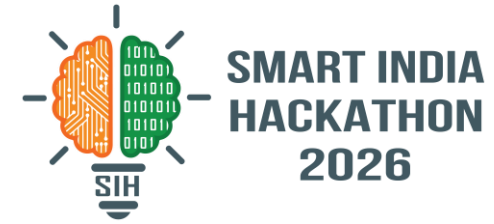
- Prototype, surface nodes: ESP-NOW to the USB gateway.
- Rover commands use a separate control path. Not the node radio.
- Not claimed: underground Wi-Fi / ESP-NOW range through rock or coal.
- Field path (after an RF survey): LoRa, mesh, repeater, leaky-feeder, or mine-approved infrastructure — future work, not this demo.

Method, every accepted reading

- Is the packet fresh, valid, and from a known device ID?
- Compute change from the captured baseline (tilt / vibration).
- Demand persistence — one tap is not an alert.
- Run rules and Isolation Forest. Never let AI override a rule.
- Show the reason. If the officer decides, send the rover and log it.
- 1 Hz firmware design. Latency and radio range: TBD — test pending.



FEASIBILITY AND VIABILITY



- Feasibility · challenges and risks · how we handle them

Build now

Two surface nodes, USB gateway, laptop dashboard, remote rover.
Isolation Forest on live baseline samples.
Local only — no cloud dependency for the demo.

Honest limit

A tabletop model is not a mine.
Tilt is orientation, not centimetres of settlement.
Green means healthy data, not a certified-safe mine.

Field path

Surveyed mounts and site-calibrated thresholds.
DGMS-approved / certified sensing and comms.
Collapse models need geotech / InSAR labels we do not have.

Risks we already design for

- False green: a missing packet can look fine. After 5 s stale → UNKNOWN, never NORMAL.
- One noisy sample (tap, wire tug): require 3 consecutive samples before Watch / Alert.
- Rover IMU mistaken for Node B: same sensor family, different job. Device ID on every reading.
- ESP32 ADC2 fights wireless: crack slider only on GPIO34 / ADC1, or drop millimetre claims.
- AI over-claim on a short tabletop set: Isolation Forest cannot clear a rule alert.
- Unlinked crack knob is not a gap: two-point ruler scale, or do not quote millimetres.
- Venue Wi-Fi / cloud demos die: laptop-local dashboard; backup video ready.
- COTS ESP32 / MPU6050 are not intrinsically safe. Tabletop only. Underground use needs DGMS-approved equipment for that hazardous area.

- Who it is for · what we can honestly claim · what comes next

Who this helps

- Mine safety officer: evidence, written reason, and history — not a serial dump.
- Inspection crew: rover goes in after Watch / Alert; nodes keep watching.
- College / SIH judges: live loop — disturb → warn → inspect → log.
- Later mine operator: same dashboard language; certified sensors can replace hobby heads.

Benefits we will actually stand behind

- Social: fewer unnecessary entries into a disturbed heading on the demo analogue.
- Economic: a standing ESP32-class watch vs waiting for a specialist survey.
- Disaster: earlier notice of surface tilt and vibration — the PS proxy for subsidence, not proof of collapse.
- User experience: colour status, charts, written reason, rover pad. Acknowledging an alert does not erase the event.

Status discipline on the dashboard

NORMAL / WATCH / ALERT / UNKNOWN. Stale or missing telemetry → UNKNOWN, never false NORMAL. Green means healthy data. It does not mean a certified-safe mine. Rover is secondary inspection, idle until requested.

NOW

Tabletop nodes + rover + local dashboard.

NEXT

More calibrated nodes, spatial view, extended validation.

FIELD

Surveyed mounts, site thresholds, DGMS-approved sensing / comms, mine monitoring integration.



RESEARCH AND REFERENCES



- References · what we will show on 10 September

Sources actually used

1. Coal Mines Regulations, 2017 · DGMS — duty to report dangerous occurrences.
2. SIH 2026 · SIH26025 · Ministry of Coal · Hardware.
3. MoC Annual Report 2024-25, Ch.14, Table 2 — CIL accident counts to Nov 2024 (not a cause-share %).
4. Liu, Ting, Zhou — Isolation Forest, ICDM 2008. Library: scikit-learn IsolationForest.
5. TDK / InvenSense MPU-6000/6050 datasheet — tilt / vibration sensor limits.
6. Espressif ESP-NOW and ESP32 ADC docs — node radio; ADC1 vs wireless wiring.
7. Sivakumar C., Srinivasan C., Gupta R.N. (2005) RaSiM 6 — Rajendra Colliery / SECL microseismic case.
8. DGMS (S&T) Tech. Circular No. 01 of 2017, cl. 6.1 — longwall geophones; not implemented here.
9. SBAS-InSAR literature — later fusion path only, not claimed as built.

If a judge has seen bigger claims on other SIH decks

- “AI predicts mine collapse %” → Isolation Forest on tabletop normal data + rule thresholds.
- “LoRa mesh across a colliery” → ESP-NOW on the tabletop. LoRa is not demonstrated.
- “Autonomous rover / AI vision” → remote drive. Camera only if shown live.
- “Green = the mine is safe” → green = healthy data, no configured trigger.